

MORNING

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Uni. Roll No.

Program/ Course: B.Tech (Batch 2018 onward)
Semester: 3rd
Name of Subject: Mathematics-III
Subject Code: BSCS-101
Paper ID: 16013
Scientific Calculator is Allowed

Time Allowed: 3 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory.
- 2) Part -C has Two questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately.

[Marks: 02 each]

Part -A

Q1.

- a) Separate $f(z) = 1/z$ into real and imaginary parts.
- b) Solve the system of equations : $2x + y = 3$, $x + 2y = 3$.
- c) Evaluate $\int_C \frac{dz}{(z+1)^2}$ where C is the circle $|z-3| = \sqrt{3}$.
- d) Write algorithm for straight line fit of a curve.
- e) Check the correctness of the statement, " Mean of a binomial distribution is 5 and variance is 9."
- f) write Cauchy Riemann equations in Cartesian form.

Part -B

[Marks: 04 each]

Q2. Fit the curve $y = ax + b$ to the following data :

x	0	1	3	6	8
y	1	3	2	5	4

Q3. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in the region $1 < |z| < 3$.

Q4. Solve the following equations by Gauss Elimination method :

$$3x + y + 2z = 3, 2x - 3y - z = -3, x + 2y + z = 4.$$

Q5. The heights of 10 males of a given locality were found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% level of significance. (Tabulated value = 1.833)

Q6. The probability that a bomb dropped from an aeroplane will strike a certain target is $1/5$. If 6 bombs are dropped, find the probability that at least two will strike the target.

Q7. If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$.

[Marks: 12 each]

Part -C

Q8. Evaluate $\int_0^\pi \frac{1}{(17 - 8 \cos \theta)} d\theta$ using contour integration.

OR

Evaluate $\int_C \frac{z^2}{(z+1)^2(z-2)} dz$, where C is $|z| = \frac{5}{2}$, using Cauchy Residue theorem.

Q9. Fit a Poisson distribution to the following data and test the goodness of fit :

x	0	1	2	3	4
f	46	38	22	9	1

(Tabulated value = 7.82)

OR

A random variable X has the following probability distribution :

X	0	1	2	3	4	5
(i) P(X)	a	2a	3a	4a	1.5a	0.5a

- Find the value of a .
- Find $P(0 < X \leq 4)$.
- The distribution function $F(x)$ of X .
